

OW Yellow Caps

Data Quality Assessment Report

Executive Insights from NYC Yellow Taxi Trips · January–March 2025

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1. Introduction

The NYC Yellow Taxi Trips dataset (January–March 2025) contains detailed records of taxi rides across New York City, capturing trip distance, fares, passenger counts, pickup and dropoff locations, and payment details. The dataset spans 11,200,280 records across three months.

The dataset and its accompanying data dictionary were provided by the course supervisor, Dr Osama Abdel Hay, as part of the OW Yellow Caps Data Visualisation consulting assignment. The taxi zone lookup table was not included in the provided assignment materials and was independently obtained from the official TLC source referenced in the Terms of Reference [3].

Before any analytical conclusions can be drawn, it is essential to evaluate the quality of the data. Poor data quality — including invalid coded values, impossible numeric entries, temporal contradictions, and missing fields — directly undermines the reliability of any business insight derived from the dataset. This report documents all identified data quality issues and explains why each constitutes an error, grounded in the official TLC data dictionary issued with the assignment.

Total records loaded: 11,200,280. The following sections address seven categories of data quality issues identified through systematic inspection of every field against the official TLC data dictionary provided in the Terms of Reference.

2. Summary of All Issues

The table below consolidates all findings across all seven quality categories. Severity is assigned as: HIGH (materially affects analysis conclusions), MEDIUM (affects specific analytical dimensions), or LOW (minor, documented only).

Check	Rows Affected	% of Total	Severity
1.1 Invalid RatecodeID	2,265,960	20.231%	HIGH
1.2 Invalid payment_type	3,415	0.030%	MEDIUM
1.3 Invalid store_and_fwd_flag	2,265,344	20.226%	MEDIUM
2.1a Negative fare_amount	535,605	4.782%	HIGH
2.1b Zero fare_amount	4,493	0.040%	MEDIUM
2.2a Negative trip_distance	1,713	0.015%	HIGH
2.2b Zero trip_distance	294,453	2.629%	HIGH
2.2c trip_distance > 100 miles	548	0.005%	MEDIUM
2.3a passenger_count = 0	69,107	0.617%	MEDIUM
2.3b passenger_count > 6	47	0.000%	LOW
2.4a Negative total_amount	186,983	1.669%	HIGH
2.4b total_amount < fare_amount	183,326	1.637%	HIGH
2.4c Zero total_amount	1,603	0.014%	MEDIUM
2.5a Negative tip_amount	360	0.003%	HIGH
2.5b tip_amount > fare_amount	548,450	4.897%	MEDIUM
2.6 mta_tax ≠ \$0.50 on standard trips	169,085	1.510%	MEDIUM
2.7 Negative tolls_amount	14,399	0.129%	MEDIUM
2.8 Invalid improvement_surcharge	183,094	1.635%	MEDIUM
2.9 Invalid congestion_surcharge	148,884	1.329%	HIGH
2.10 Invalid airport_fee values	33,774	0.302%	HIGH
2.11 Invalid cbd_congestion_fee values	91,117	0.813%	HIGH
2.12 Negative extra values	92,939	0.830%	MEDIUM
3.1 Dropoff before pickup	2,034	0.018%	HIGH
3.2 Zero-duration trips	29,201	0.261%	MEDIUM
3.3a Duration > 3 hours	4,252	0.038%	MEDIUM
3.3b Duration > 5 hours	3,810	0.034%	MEDIUM
3.4 Dates outside Jan–Mar 2025	25	0.000%	HIGH
4.1 Cash trips with tip_amount > 0	2,913	0.026%	HIGH
4.2 airport_fee on non-airport pickups	26,597	0.237%	MEDIUM
4.3 cbd_congestion_fee before Jan 5	2,160	0.019%	HIGH
4.4 total_amount arithmetic mismatch	4,335,445	38.708%	HIGH
4.5 Zero distance with positive fare	244,747	2.185%	HIGH
4.6 extra > fare_amount	556,443	4.968%	MEDIUM
5.2a Unknown PU zone (264/265)	27,397	0.245%	MEDIUM

5.2b Unknown DO zone (264/265)	69,553	0.621%	MEDIUM
6.1 Missing passenger_count	2,267,572	20.246%	HIGH
6.1 Missing RatecodeID	2,264,213	20.216%	HIGH
6.1 Missing congestion_surcharge	2,264,213	20.216%	HIGH
6.1 Missing payment_type	1,702	0.015%	LOW
7.2 Near-duplicate trips	32,537	0.291%	MEDIUM

3. Field Validity — Out-of-Range Coded Values

The TLC data dictionary defines exact valid values for four coded fields. Any value outside the defined set is considered invalid, regardless of how numerically close it is to a valid code.

3.1 RatecodeID

Valid values per the TOR data dictionary: 1, 2, 3, 4, 5, 6, and 99.

Rows affected	2,265,960 (20.231% of total)
Severity	HIGH

Why this is an error

The data dictionary explicitly lists the only permissible values for RatecodeID. Values of -1.0, 12.0, and 88.0 do not correspond to any defined rate category. A RatecodeID of -1 is mathematically impossible as a category code. Values 12 and 88 have no defined meaning in the TLC system. The volume of invalid entries — over 20% of the dataset — is primarily driven by 2,264,213 NaN (null) records, which indicate the rate code was never recorded by the vendor.

RatecodeID 1.0 (Standard)	8,391,091
RatecodeID NaN (missing)	2,264,213
RatecodeID 2.0 (JFK)	288,617
RatecodeID 99.0 (Null/unknown)	125,079
RatecodeID 5.0 (Negotiated)	81,028
RatecodeID 3.0 (Newark)	27,176
RatecodeID 4.0 (Nassau/Westchester)	21,318
RatecodeID -1.0 (INVALID)	607
RatecodeID 12.0 (INVALID)	607
RatecodeID 88.0 (INVALID)	533
RatecodeID 6.0 (Group ride)	11

3.2 payment_type

Valid values per the TOR data dictionary: 0, 1, 2, 3, 4, 5, and 6.

Rows affected	3,415 (0.030% of total)
Severity	MEDIUM

Why this is an error

The data dictionary assigns a defined meaning to each payment_type code. Values 7, 8, and 9 are not defined anywhere in the TLC documentation and therefore have no interpretable business meaning. These records cannot be classified by payment method and must be excluded from any analysis of payment behavior, tipping rates, or revenue split by payment channel.

payment_type 1 (Credit card)	7,483,854
payment_type 0 (Flex Fare)	2,263,735
payment_type 2 (Cash)	1,135,571
payment_type 4 (Dispute)	241,212
payment_type 3 (No charge)	72,492
payment_type 7 (INVALID)	549
payment_type 8 (INVALID)	605
payment_type 9 (INVALID)	559
payment_type NaN (INVALID)	1,702

3.3 store_and_fwd_flag

Valid values per the TOR data dictionary: "Y" (store and forward trip) or "N" (not a store and forward trip).

Rows affected	2,265,344 (20.226% of total)
Severity	MEDIUM

Why this is an error

The field is a binary flag with only two defined states. Values of "X", "?", "Z", and None fall outside this definition. The characters X, ?, and Z carry no defined meaning in the TLC system. The None values (2,264,213) correspond to the same vendor-driven block of missing data identified in RatecodeID and are likely records submitted without vehicle memory connectivity data. While this field is not central to fare or demand analysis, its mass invalidity reinforces the finding that VendorID 6 submitted structurally incomplete records.

N (not store and forward)	8,891,905
None (missing)	2,264,213
Y (store and forward)	43,031
X (INVALID)	419

? (INVALID)	356
Z (INVALID)	356

4. Numeric Range Violations

Several numeric fields contain values that are physically or logically impossible. A negative fare, a negative distance, or a negative tip cannot occur in any valid taxi trip. These are not outliers — they are data errors.

4.1 fare_amount

Negative fare_amount (< 0)	535,605 rows — 4.782% → Exclude
Zero fare_amount (= 0)	4,493 rows — 0.040% → Flag
Minimum recorded value	-1,807.60
Maximum recorded value	863,372.12
Mean	\$17.24

Why this is an error

A fare_amount below zero is impossible under the TLC metered fare system. The meter calculates the fare based on elapsed time and distance — it cannot produce a negative result. A value of -1,807.60 is not a rounding artifact; it represents a fundamental recording failure. These records cannot be used in any fare-based analysis. Zero-fare records are flagged separately as they may indicate cancelled trips or testing entries where the meter was never properly engaged.

4.2 trip_distance

Negative trip_distance (< 0)	1,713 rows — 0.015% → Exclude
Zero trip_distance (= 0)	294,453 rows — 2.629% → Flag
trip_distance > 100 miles	548 rows — 0.005% → Flag
Minimum recorded value	-38.09 miles
Maximum recorded value	320,136.29 miles
Mean	6.18 miles

Why this is an error

The taximeter measures elapsed distance as a non-negative physical quantity. A negative distance of -38.09 miles has no physical interpretation and indicates a recording or transmission error. A maximum of 320,136.29 miles — equivalent to over 12 trips around the Earth — is clearly impossible for a single NYC taxi trip and represents a catastrophic data entry failure. Zero-distance trips where the duration is also non-zero may indicate meter activation without vehicle movement, but cannot be distinguished from test activations without additional context.

4.3 passenger_count

passenger_count = 0	69,107 rows — 0.617% → Flag
passenger_count > 6	47 rows — 0.000% → Flag
Missing (NaN)	2,267,572 rows — 20.246%

Why this is an error

NYC taxi regulations cap the legal passenger limit at six. Any record reporting more than six passengers violates this constraint and is therefore invalid. A passenger count of zero is physically ambiguous — a trip with no passenger may indicate a deadhead trip, a test activation, or a driver error in entering the count. These cannot be treated as normal revenue trips. The 2.26 million missing values are addressed in the Missingness section.

4.4 total_amount

Negative total_amount (< 0)	186,983 rows — 1.669% → Exclude
total_amount < fare_amount	183,326 rows — 1.637% → Flag
Zero total_amount (= 0)	1,603 rows — 0.014% → Flag
Minimum recorded value	-1,832.85
Maximum recorded value	863,380.37

Why this is an error

The total_amount field represents the total charge to the passenger. Per the data dictionary, total_amount is the sum of all itemized components (fare, taxes, surcharges, tolls, and tips for credit card payments). A negative total is impossible — the passenger cannot owe a negative amount to the driver. Additionally, total_amount must always be greater than or equal to fare_amount because it includes the fare plus additional charges. A total below the base fare indicates an arithmetic inconsistency in the submission.

4.5 tip_amount

Negative tip_amount (< 0)	360 rows — 0.003% → Exclude
tip_amount > fare_amount	548,450 rows — 4.897% → Flag
Minimum recorded value	-220.00
Maximum recorded value	440.00
Mean	\$2.85

Why this is an error

A negative tip is impossible. The tip_amount field is, per the data dictionary, automatically populated for credit card tips only — it is not manually entered by the driver. A negative auto-populated tip indicates a system-level recording error. The 548,450 cases where tip_amount exceeds fare_amount are flagged as extreme outliers — while a passenger can legally tip more than the fare, values this numerous warrant scrutiny and may include misclassified amounts.

4.6 mta_tax

mta_tax ≠ \$0.50 on standard rate trips	169,085 rows — 1.510% → Document
mta_tax = -0.50 (negative)	163,181 rows
mta_tax = 0.00 on standard trips	5,892 rows
mta_tax > \$0.50 on standard trips	201 rows

Why this is an error

The TLC imposes a fixed \$0.50 MTA tax on all metered trips under RatecodeID 1 (Standard rate). This is not a variable charge — it is a statutory fixed amount. Any standard-rate trip showing mta_tax ≠ \$0.50 is inconsistent with the regulated fare structure. The 163,181 records with mta_tax = -0.50 are particularly notable: a negative tax is not legally possible and likely represents a refund or reversal entry that was incorrectly recorded in the trip submission rather than as a separate adjustment.

4.7 tolls_amount

Negative tolls_amount (< 0)	14,399 rows — 0.129%
Severity	MEDIUM
Example invalid value	-6.94

Why this is an error

Tolls are fixed charges applied by toll authorities — they cannot be negative. A negative tolls_amount has no physical or financial interpretation; a toll cannot be refunded through the trip record system. These records indicate either a reversal entry incorrectly captured as a trip field or a system recording failure.

4.8 improvement_surcharge

Invalid improvement_surcharge values	183,094 rows
Severity	MEDIUM
Invalid value observed	-1.0

Why this is an error

The improvement surcharge is a fixed statutory charge of \$0.30 per trip, introduced in 2015 to fund taxi accessibility improvements. It is assessed at the flag drop and cannot be negative. A value of -1.0 is both the wrong magnitude and the wrong sign — it has no valid interpretation under the TLC fare structure and indicates a recording error in the surcharge field.

4.9 congestion_surcharge

Expected valid values	2.50 or 0.00
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Negative congestion_surcharge (-2.50)	148,883 rows — 1.329%
Invalid value (1.0)	1 row
Missing (NaN)	2,264,213 rows — 20.216%
Severity	HIGH

Why this is an error

The NYS congestion surcharge is a fixed charge with only two defined states: .50 (applied) or /bin/sh.00 (not applied). A value of -2.50 is impossible — the surcharge cannot be negated through the trip record. A value of 1.0 does not correspond to any defined surcharge amount. These entries are internally contradictory with the regulatory fare structure and cannot be used in any surcharge or revenue analysis.

4.10 airport_fee

Expected valid values	0.00, 1.25, or 1.75
Negative airport_fee values	32,944 rows
airport_fee = 6.75 (INVALID)	645 rows
airport_fee = 5.00 (INVALID)	184 rows
airport_fee = 0.75 (INVALID)	1 row
Missing (NaN)	2,263,828 rows — 20.212%
Severity	HIGH

Why this is an error

The airport_fee field has only three defined valid values: /bin/sh.00 (no airport pickup), .25 (LaGuardia), and .75 (JFK). Values of 6.75, 5.00, and 0.75 do not correspond to any defined airport surcharge in the TLC fare schedule. Negative values are impossible — an airport fee cannot be refunded through the trip record. These entries violate the explicit fare structure and cannot be used in airport-related revenue analysis.

4.11 cbd_congestion_fee — invalid values

Expected valid values	0.75 (standard) or 0.00 (not applicable)
Negative cbd_congestion_fee values	90,978 rows
cbd_congestion_fee = 1.25 (INVALID)	129 rows
cbd_congestion_fee = 1.50 (INVALID)	9 rows
cbd_congestion_fee = 0.25 (INVALID)	1 row
Non-zero before Jan 5, 2025	2,160 rows (covered in Section 6.3)
Severity	HIGH

Why this is an error

The `cbd_congestion_fee` has only two defined valid states: `/bin/sh.75` (the standard per-trip Congestion Relief Zone charge) or `/bin/sh.00` (not applicable). Values of 1.25, 1.50, and 0.25 do not correspond to any defined congestion fee tier. Negative values are impossible — the charge cannot be reversed through the trip record system. These entries are inconsistent with the MTA congestion pricing framework and must be excluded from any congestion fee analysis.

4.12 extra

Negative extra values	92,939 rows — 0.830%
Irregular values observed	0.37, 0.26, 8.80 and others
Severity	MEDIUM

Why this is an error

The `extra` field captures miscellaneous extras and surcharges — a combination of applicable charges that should always be non-negative. Negative values such as -0.05 and -0.07 are impossible as surcharge amounts and indicate recording errors. Additionally, rare values such as 0.37, 0.26, and 8.80 do not correspond to standard NYC taxi surcharge denominations, suggesting either data entry errors or undocumented charge categories that cannot be verified against the TLC fare schedule.

5. Outlier Detection

Outlier analysis using the 99th percentile threshold identifies the top 1% of values as unusually high across key numeric fields. While not all outliers are errors, they affect average-based metrics and must be disclosed.

Column	P99 Threshold	Outlier %
<code>trip_distance</code>	19.34 miles	1.00%
<code>fare_amount</code>	1.30	1.00%
<code>total_amount</code>	02.11	0.99%
<code>tip_amount</code>	7.05	1.00%
<code>tolls_amount</code>	.94	0.41%
<code>extra</code>	.50	0.64%
<code>mta_tax</code>	<code>/bin/sh.50</code>	0.002%
<code>improvement_surcharge</code>	.00	0.00%
<code>congestion_surcharge</code>	.50	0.00%
<code>Airport_fee</code>	.75	0.007%

The main concern is the `extra` column, which shows outliers alongside negative and irregular values — indicating a broader data quality problem beyond simple extreme values. The `mta_tax`, `improvement_surcharge`, and `congestion_surcharge` columns show near-zero outlier rates, consistent with their fixed statutory amounts. Fields such as `trip_distance` and `fare_amount` show 1% outlier rates that likely represent legitimate long or expensive trips rather than errors.

6. Temporal Consistency

Every trip must obey basic temporal logic: the dropoff must occur after the pickup, and all records must fall within the stated dataset scope of January through March 2025.

6.1 Dropoff before pickup

Rows affected	2,034 (0.018% of total)
Severity	HIGH
Example	trip_id 3816: pickup 2025-01-01 00:30 → dropoff 2024-12-31 23:20

Why this is an error

A taxi trip cannot end before it begins. The `tpep_pickup_datetime` marks when the meter was engaged and `tpep_dropoff_datetime` marks when it was disengaged — these are sequential physical events. A dropoff timestamp that precedes the pickup timestamp by up to 70 minutes (as seen in the sample) cannot represent any real trip. These records cannot be used in any temporal, duration, or demand analysis.

6.2 Zero-duration trips

Rows affected	29,201 (0.261% of total)
Severity	MEDIUM

Why this is an error

A trip where pickup and dropoff timestamps are identical has a computed duration of zero seconds. No billable trip can have zero duration — the meter requires time to engage, record movement, and disengage. These records either represent meter test activations, immediately cancelled trips where the meter was engaged and then disengaged without movement, or submission errors. They cannot be treated as valid billable trips.

6.3 Extremely long trips

Duration > 3 hours	4,252 rows — 0.038% → Flag
Duration > 5 hours	3,810 rows — 0.034% → Flag
Maximum trip duration	8,564.67 minutes (nearly 6 days)

Why this is an error

While some legitimate NYC taxi trips may exceed three hours (e.g. traffic-delayed airport runs), a maximum recorded duration of 8,564 minutes — equivalent to nearly six full days — is impossible for any real trip. The legal maximum NYC taxi fare cap also limits the practical duration of metered trips. These extreme values indicate cases where the meter was left running after the passenger exited, the vehicle was impounded or taken out of service without disengaging the meter, or a timestamp recording failure occurred.

6.4 Dates outside Jan–Mar 2025 scope

Rows affected	25 (0.000% of total)
Severity	HIGH
Earliest date found	2007-12-05 18:45:00
Latest date found	2025-04-01 00:00:17

Why this is an error

The dataset is defined as covering January through March 2025 exclusively. A pickup timestamp of 2007-12-05 is 17 years outside the dataset scope and cannot represent a 2025 trip — it is either a system clock error or a corrupted record. The April 2025 record may represent a genuine trip that crossed midnight on March 31 but was submitted with rounding error. All records outside the defined scope are excluded to maintain analytical integrity.

7. Cross-Field Logical Consistency

Several fields have defined relationships with other fields. Where these relationships are violated, the record contains an internal contradiction that makes at least one of the fields unreliable.

7.1 Cash trips with non-zero tip_amount

Rows affected	2,913 (0.026% of total)
Severity	HIGH

Why this is an error

The data dictionary explicitly states: "tip_amount — This field is automatically populated for credit card tips. Cash tips are not included." This is a system-level constraint, not a preference. When payment_type = 2 (Cash), the system does not record tip_amount — it is technically impossible for the meter system to auto-populate a tip on a cash transaction. Any non-zero tip_amount on a cash trip therefore cannot have been recorded legitimately and represents either a data entry override, a system glitch, or a misclassified payment type. These records corrupt any analysis of tipping behavior by payment method.

7.2 airport_fee on non-airport pickups

Rows affected	26,597 (0.237% of total)
Severity	MEDIUM
Most common invalid PU zone	Zone 70 — 23,387 occurrences

Why this is an error

Per the data dictionary, airport_fee applies "for pick up only at LaGuardia and John F. Kennedy Airports" — zones 132 and 138 respectively. An airport_fee charged on a pickup from any other zone violates this rule. Zone 70 (which accounts for 23,387 of the flagged records) is not an airport zone; charging an airport_fee there constitutes either a billing system error or a fare misclassification. These records affect the reliability of fare component analysis and geographic revenue breakdowns.

7.3 cbd_congestion_fee before January 5, 2025

Rows affected	2,160 (0.019% of total)
Severity	HIGH
Source of the Jan 5 cutoff	TOR Appendix A and data dictionary (page 5)
Mean CBD fee before Jan 5	\$0.0044 (should be \$0.00)
Mean CBD fee from Jan 5 onward	\$0.5377

Why this is an error

The Terms of Reference state explicitly in Appendix A: "cbd_congestion_fee begins on January 5, 2025." The data dictionary confirms this: "Per-trip charge for MTA's Congestion Relief Zone starting Jan. 5, 2025." Any non-zero cbd_congestion_fee on a trip with a pickup before January 5, 2025 is therefore impossible under the legal framework — the charge did not exist yet. These 2,160 records were either submitted in error, represent test records, or carry a fee from a system that was incorrectly activated before the official start date. They must be excluded from any analysis of congestion pricing impact.

7.4 total_amount arithmetic mismatch

Rows affected	4,335,445 (38.708% of total)
Severity	HIGH
Tolerance applied	\$0.10 (to allow for rounding)
Mean absolute difference	\$1.14
Maximum absolute difference	\$102.50

Why this is an error

Per the data dictionary, total_amount is defined as the sum of: fare_amount + extra + mta_tax + tip_amount + tolls_amount + improvement_surcharge + congestion_surcharge + airport_fee + cbd_congestion_fee. A \$0.10 tolerance is applied to accommodate rounding at the cent level. Any gap larger than \$0.10 between total_amount and the computed sum of its components indicates that either a fee component was omitted from the record, an undocumented charge was included in total_amount, or the total was recorded incorrectly. With 38.7% of all records affected and a mean gap of \$1.14, this is the most pervasive arithmetic quality issue in the dataset.

7.5 Zero trip_distance with positive fare_amount

Rows affected	244,747 (2.185% of total)
Severity	HIGH

Why this is an error

A trip that covers zero distance cannot generate a time-and-distance fare under the standard rate. The TLC fare structure charges a base fare at flag drop plus a per-mile and per-minute rate — a stationary vehicle with the meter running would accumulate time charges but not distance charges, making a substantial fare_amount on zero distance logically contradictory. These 244,747 records represent either meter activations without vehicle

movement, GPS failures causing distance to not be recorded, or test entries. They cannot be used in any analysis linking fare revenue to geographic movement.

7.6 extra greater than fare_amount

Rows affected	556,443 (4.968% of total)
Severity	MEDIUM

Why this is an error

The extra field captures miscellaneous surcharges — peak hour charges, overnight fees, and similar additions. While extra can legitimately exceed the base fare for very short trips (e.g. a brief trip during rush hour), 556,443 such cases (4.97% of all records) is an unusually high volume. In the context of records that also carry other anomalies (negative fares, zero distances), this cross-field pattern reinforces that a subset of records contains systematically distorted financial data rather than legitimate short-trip economics.

8. Location Validity

Pickup and dropoff location IDs are validated against the official TLC Taxi Zone Lookup Table, downloaded directly from the TLC trip record data page (the authoritative source referenced in TOR reference [3]).

8.1 All LocationIDs within valid range

Invalid PULocationID	0 rows — fully clean
Invalid DOLocationID	0 rows — fully clean
Zone file source	https://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv
Valid zones loaded	265

All PULocationID and DOLocationID values in the dataset are present in the official TLC zone lookup table. No records use zone IDs outside the defined set of 265 zones. This check passes completely.

8.2 Unknown location codes (zones 264 and 265)

PULocationID = 264/265	27,397 rows — 0.245%
DOLocationID = 264/265	69,553 rows — 0.621%

Why this requires attention

Zones 264 and 265 are valid TLC codes — they appear in the official lookup table and represent "Unknown" and "N/A" location designations. They are not data entry errors. However, these codes indicate that the taximeter could not resolve the geographic location of the pickup or dropoff. Records with these codes cannot be used in any borough-level or zone-level geographic analysis, as their location is definitionally unknown. The fact that unknown dropoff zones (0.621%) are nearly three times more frequent than unknown pickup zones (0.245%) suggests drivers are more likely to be in unresolvable locations at trip end than at trip start.

9. Missingness

Missing values were counted across all columns. The pattern of missingness — specifically its concentration in a single vendor — is the most operationally significant finding in this category.

9.1 Null counts by column

passenger_count	2,267,572 missing — 20.246%
RatecodeID	2,264,213 missing — 20.216%
store_and_fwd_flag	2,264,213 missing — 20.216%
congestion_surcharge	2,264,213 missing — 20.216%
Airport_fee	2,263,828 missing — 20.212%
payment_type	1,702 missing — 0.015%

9.2 Missingness concentrated in VendorID 6

VendorID 1 — passenger_count null rate	14.199%
VendorID 2 — passenger_count null rate	21.902%
VendorID 6 — passenger_count null rate	100.000%
VendorID 7 — passenger_count null rate	0.033%

Why this is an error

VendorID 6 (Myle Technologies Inc) submitted 100% null values for `passenger_count`, `RatecodeID`, `store_and_fwd_flag`, and `congestion_surcharge`. This is not random missingness — it is a complete, systematic failure to populate these fields across every single record from that vendor. This means that for all trips provided by VendorID 6, four key analytical dimensions are entirely unavailable. Any analysis involving these fields must either exclude VendorID 6 records entirely or treat VendorID 6 as a separate analytical segment with acknowledged data limitations. This vendor-specific gap also explains why the overall null rate for these columns is approximately 20% — it directly tracks the proportion of records from VendorID 6.

10. Duplicates

10.1 Exact duplicate trip_id

Rows affected	0 (0.000% of total)
Severity	None — check passed

No exact duplicate `trip_id` values were found. The proxy row identifier is unique across all 11,200,280 records.

10.2 Near-duplicate trips

Rows affected	32,537 (0.291% of total)
Severity	MEDIUM
Definition	Same tpep_pickup_datetime + PULocationID + fare_amount

Why this requires attention

A near-duplicate is defined as two or more records sharing an identical pickup timestamp, pickup zone, and fare amount. While it is theoretically possible for two different taxis to pick up passengers at the exact same second in the same zone for the exact same fare, this is extremely unlikely at scale. The sample data shows trip_id 4532 and 4533 sharing pickup time 2025-01-01 00:01:19 at zone 164 with fare \$0.00 but with total_amounts of -2.50 and +2.50 respectively — a pattern consistent with a correction entry being submitted alongside the original. These 32,537 records may inflate trip counts and revenue totals if both copies are retained.

11. Conclusion

The NYC Yellow Taxi dataset for January–March 2025 contains 11,200,280 records across which multiple categories of data quality issues were identified. The most operationally significant findings are:

- VendorID 6 submitted records with 100% null values across four key fields, affecting approximately 20% of the entire dataset.
- 38.7% of records show a discrepancy between total_amount and the sum of its fee components, the single largest arithmetic quality failure.
- 4.782% of records carry a negative fare_amount, which is physically impossible under the metered fare system.
- 20.231% of RatecodeID values are invalid, primarily due to the VendorID 6 null block combined with undocumented codes (-1, 12, 88).
- The cbd_congestion_fee pre-dates its legally mandated start of January 5, 2025 in 2,160 records.

All findings are documented with their row counts, percentages of total records, and specific reasons grounded in the TLC data dictionary and the Terms of Reference. Detailed cleaning decisions and sensitivity analysis are provided in the accompanying Technical Preprocessing Report.

References

- [1] NYC Taxi and Limousine Commission. TLC Trip Record Data. <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>
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- [3] NYC TLC. Taxi Zone Lookup Table. https://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv